Baselines and fair comparisons

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Reproducible AI Research Study

THE CONNECTED WEEK

One complete baseline and comparison-condition record.

Keep procedures, cases, permitted sources, scoring, hidden-label status, budgets and evidence gaps together.

Keep one habit: Recall → W1–W3/W5 → independent time-budget case → fair-comparison project → delayed return.

MAIN LESSON · SECTION 1

Recall · Retrieve before notes

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.
- Without opening the old lesson, explain one key idea from Hypotheses and measurable outcomes. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Replication and reproduction. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Reading a research claim. Give an example and a mistake that the idea helps you avoid.

Hypotheses and measurable outcomes

My remembered idea, example and correction

Replication and reproduction

My remembered idea, example and correction

Reading a research claim

My remembered idea, example and correction

W1 · Explain the current example

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Fair baselines are credible documented alternatives under the intended task and resource conditions. For room reporting, use manual authoritative lookup or deterministic matching of the requested room, not arbitrary random names. Main assistant/current register versus baseline/obsolete register confounds information freshness with method.

  Added same-function clubs V/W/X fixture has current E/B/D versus old C/B/A:1/3 old versus 3/3 current; align both current and both 3/3. This is a constructed counterexample, not a real model benchmark or equality claim. Separate picture same eight fixed cases and permitted sources gives A 6/8=75%, B 7/8=87.5%, difference 12.5 percentage points/one net correct; paired flips unknown.

  No test-case tuning in original scenario; exposing B to hidden evaluation labels breaks unseen comparison. Permitted source facts and hidden judge answers are distinct. Resource policy/timing not supplied, so no efficiency/budget claim. These aggregate results alone do not establish stable advantage or a reported significance result.

  Preserve conditions and complete-procedure trade-offs rather than inferring universal superiority.

- Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

My reasoning and evidence

W2 · Trace the worked case

> Keep the supplied inputs, comparison rule and evidence together.

Complete task and evidence

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  Preserve conditions and complete-procedure trade-offs rather than inferring universal superiority.

- Assistant gets current register; **baseline** gets an obsolete one. Assistant scores higher. Show the intermediate steps and give the conclusion with its scope.

My reasoning and evidence

W3 · Repair a misconception

> Keep this task’s inputs,rule and response together.

Complete task and evidence

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  Added same-function clubs V/W/X fixture has current E/B/D versus old C/B/A:1/3 old versus 3/3 current; align both current and both 3/3. This is a constructed counterexample, not a real model benchmark or equality claim. Separate picture same eight fixed cases and permitted sources gives A 6/8=75%, B 7/8=87.5%, difference 12.5 percentage points/one net correct; paired flips unknown.

  No test-case tuning in original scenario; exposing B to hidden evaluation labels breaks unseen comparison. Permitted source facts and hidden judge answers are distinct. Resource policy/timing not supplied, so no efficiency/budget claim. These aggregate results alone do not establish stable advantage or a reported significance result.

  Preserve conditions and complete-procedure trade-offs rather than inferring universal superiority.

- A report omits this qualification: A **baseline** is credible relative to the task, not merely a procedure easy to outperform. Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

My reasoning and evidence

MAIN LESSON · SECTION 9

W5 · Explain the limitation

> Keep this task’s inputs,rule and response together.

Complete task and evidence

- Fair baselines are credible documented alternatives under the intended task and resource conditions. For room reporting, use manual authoritative lookup or deterministic matching of the requested room, not arbitrary random names. Main assistant/current register versus baseline/obsolete register confounds information freshness with method.

  Added same-function clubs V/W/X fixture has current E/B/D versus old C/B/A:1/3 old versus 3/3 current; align both current and both 3/3. This is a constructed counterexample, not a real model benchmark or equality claim. Separate picture same eight fixed cases and permitted sources gives A 6/8=75%, B 7/8=87.5%, difference 12.5 percentage points/one net correct; paired flips unknown.

  No test-case tuning in original scenario; exposing B to hidden evaluation labels breaks unseen comparison. Permitted source facts and hidden judge answers are distinct. Resource policy/timing not supplied, so no efficiency/budget claim. These aggregate results alone do not establish stable advantage or a reported significance result.

  Preserve conditions and complete-procedure trade-offs rather than inferring universal superiority.

- For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

  Do not reuse W4 as a new final test after receiving feedback.

My reasoning and evidence

MAIN LESSON · SECTION 4

Try alone · Selected transfer

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Model uses 12 seconds and **baseline** 2 on each case. Task has a 3-second budget. How should results be framed?

- Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Independent case: Apply a new case independently

My independent explanation, trace and limits

MAIN LESSON · SECTION 10

Project · Keep the research evidence

> Preserve the question, protocol, observations, analysis and limits.

Complete task and evidence

- Add a labelled **baseline**s and fair comparisons **evidence** sheet to Reproducible AI Research Study. Show how you can explain the **mechanism** and terminology of **baseline**s and fair comparisons.

  Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

> **My project change, supporting evidence and remaining limit**

MAIN LESSON · SECTION 11

Feedback · Repair and return

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Supported W2 record**

REFERENCE · SECTION 2

Reference · Supported W2 record

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Use W2: Trace the worked case for **Baseline**s and fair comparisons. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

The step I completed without help, and the hint I needed

▼ Extra practice · Individual reasoning check

REFERENCE · SECTION 3

Extra practice · Individual reasoning check

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Use the worked case for **Baseline**s and fair comparisons. Proposed explanation: "The worked result establishes every future case". Decide whether this explanation is supported.

  Point to the step or supplied fact that decides; explain your repair if needed.

> **My explanation before partner discussion**

▼ Reference · Earlier W4 response copy

REFERENCE · SECTION 8

Reference · Earlier W4 response copy

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Model uses 12 seconds and **baseline** 2 on each case. Task has a 3-second budget. How should results be framed?

My reasoning and evidence

▼ **Extra practice · Pictured case**

REFERENCE · SECTION 12

Extra practice · Pictured case

> Keep this task's inputs,rule and response together.

Complete task and evidence

- A **baseline** deserves a fair chance. Two classifiers answer the same eight fixed questions with the same allowed sources.

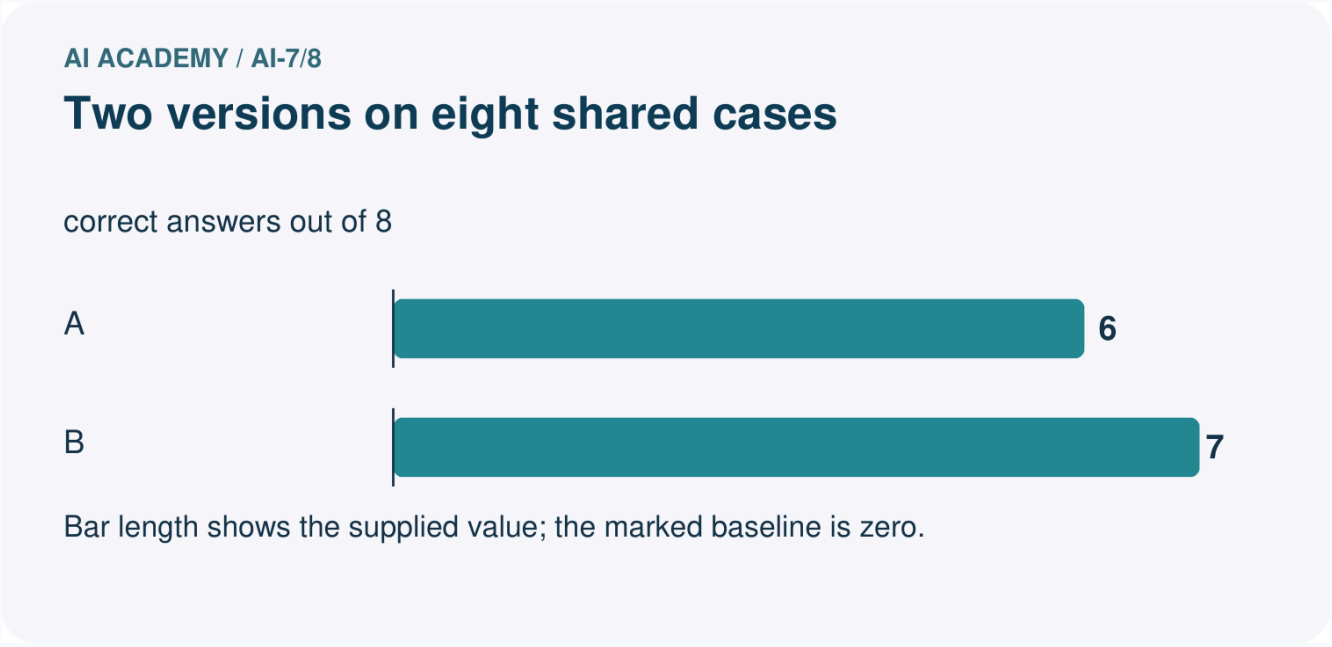
- Simple rule A: 6/8 correct. New model B: 7/8 correct. No test case is used to tune either version.

- Calculate 6/8 and 7/8 correctness and name the one-case gain. Why is giving B the answer key incompatible with an unseen comparison to the simple **baseline**?

- Support used: alone / hint / worked example. This record is practice **evidence**.

- Project connection: For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning.

  Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.



Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-7/8

Baselines and fair comparisons

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Declare baseline

A uses a simple documented rule rather than an intentionally weak placeholder.



2. Measure fairly

A is 75%; B is 87.5% on the same eight cases.



3. Interpret cautiously

B adds one correct case here; compare resources and failure types too.

CHANGE ONE THING

Give B the answer key before testing.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

My picture-based explanation and changed-case prediction

REFERENCE · SECTION 13

Reference · Project milestone

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Project: Reproducible AI Research Study
- Current milestone: Make a transparent first version. Keep a trace of how one **input** becomes an **output**; explain the rule or learned component.
- For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	
One change since the previous checkpoint and why	
One error or limit and the next test	

Explain what this week adds to your project. Reuse last week's record rather than starting again.

▼ Extra practice · Predict,check and repair

REFERENCE · SECTION 14

Extra practice · Predict,check and repair

> Keep supported feedback separate from the independent case.

First supported question and optional hint

- A **baseline** deserves a fair chance. Two classifiers answer the same eight fixed questions with the same allowed sources.
- Which labelled step matches this detail? B adds one correct case here; compare resources and failure types too.
- 1. Declare **baseline**
- 2. Interpret cautiously
- 3. Measure fairly
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check after the first attempt**

After your attempt, compare with the supported explanation: Interpret cautiously B adds one correct case here; compare resources and failure types too. A **baseline** anchors the claim of improvement.

Fair comparison keeps information access, cases and scoring aligned while allowing the declared method difference.

Changed-case question and optional hint

- Changed-case practice: Give B the answer key before testing. Which conclusion is supported by the supplied facts?
- 1. We can decide the result without examining the changed **input** or the procedure.
- 2. The resulting comparison is invalid as unseen evaluation.
- 3. This one example guarantees correct results for every future **input**.
- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check after the changed attempt**

After your attempt, compare with the supported explanation: The resulting comparison is invalid as unseen evaluation. B would receive privileged information that A did not receive.

Repair and retain the record

Review Hypotheses and measurable outcomes (ai-7/7). Goal: Explain the **mechanism** and terminology of hypotheses and measurable outcomes. Short practice: Name the predicted outcome and how unsupported is scored.

If B sounds friendlier but ties A's unsupported count, does the recorded improvement hypothesis receive support?. Return to this week and explain what you changed.

Review Replication and reproduction (ai-7/5). Goal: Explain the **mechanism** and terminology of replication and reproduction. Short practice: Calculate the scores from the original and new four-record sets.

Which activity reproduces saved analysis, which repeats the protocol, and why does a different rerun of the original data need investigation?. Return to this week and explain what you changed.

Review Reading a research claim (ai-7/2). Goal: Explain the **mechanism** and terminology of reading a research claim. Short practice: Calculate the **baseline** and new rates for 2/10 and 4/10.

Explain both twice the rate and the percentage-point change, and say why neither measures every learner's improvement.. Return to this week and explain what you changed.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test

▼ **Earlier wording, alternatives and picture transcripts**

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Keep your completed work

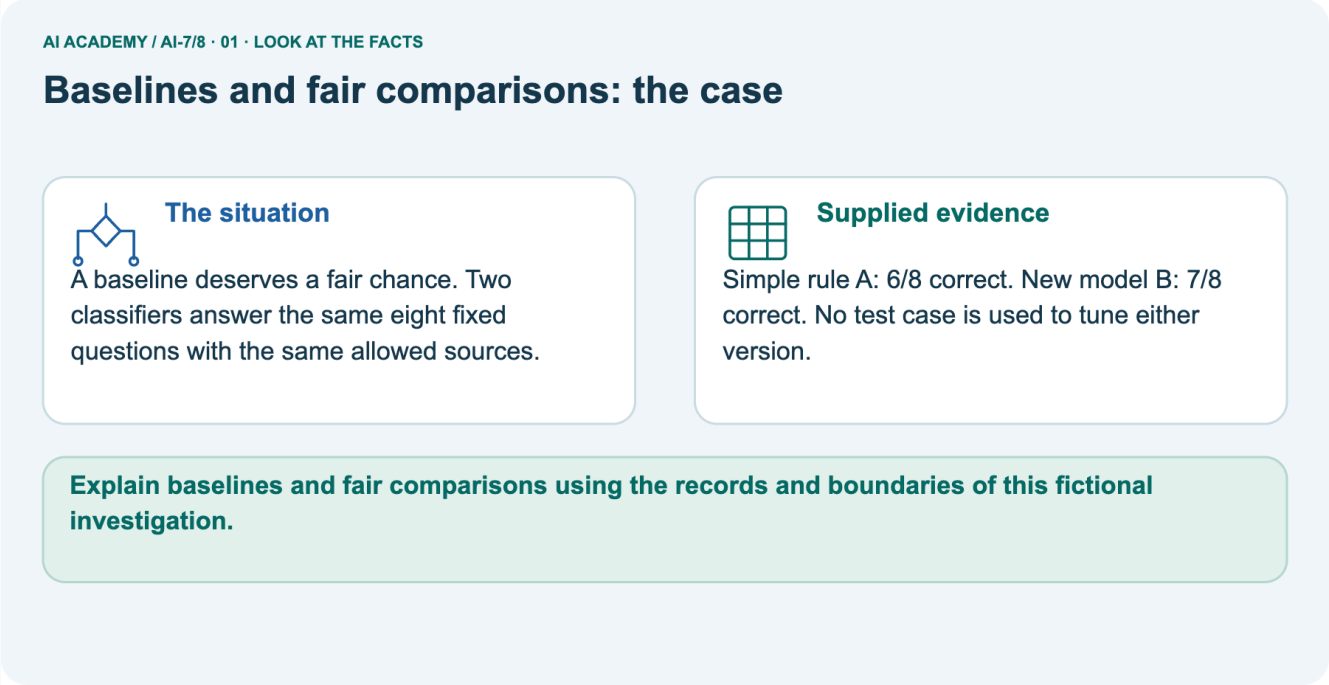
▼ **Reference · Six-picture response sheets**

REFERENCE · SECTION 15

Reference · Six-picture response sheets

> Choose the picture that helps explain the current step.

01. Baselines and fair comparisons: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

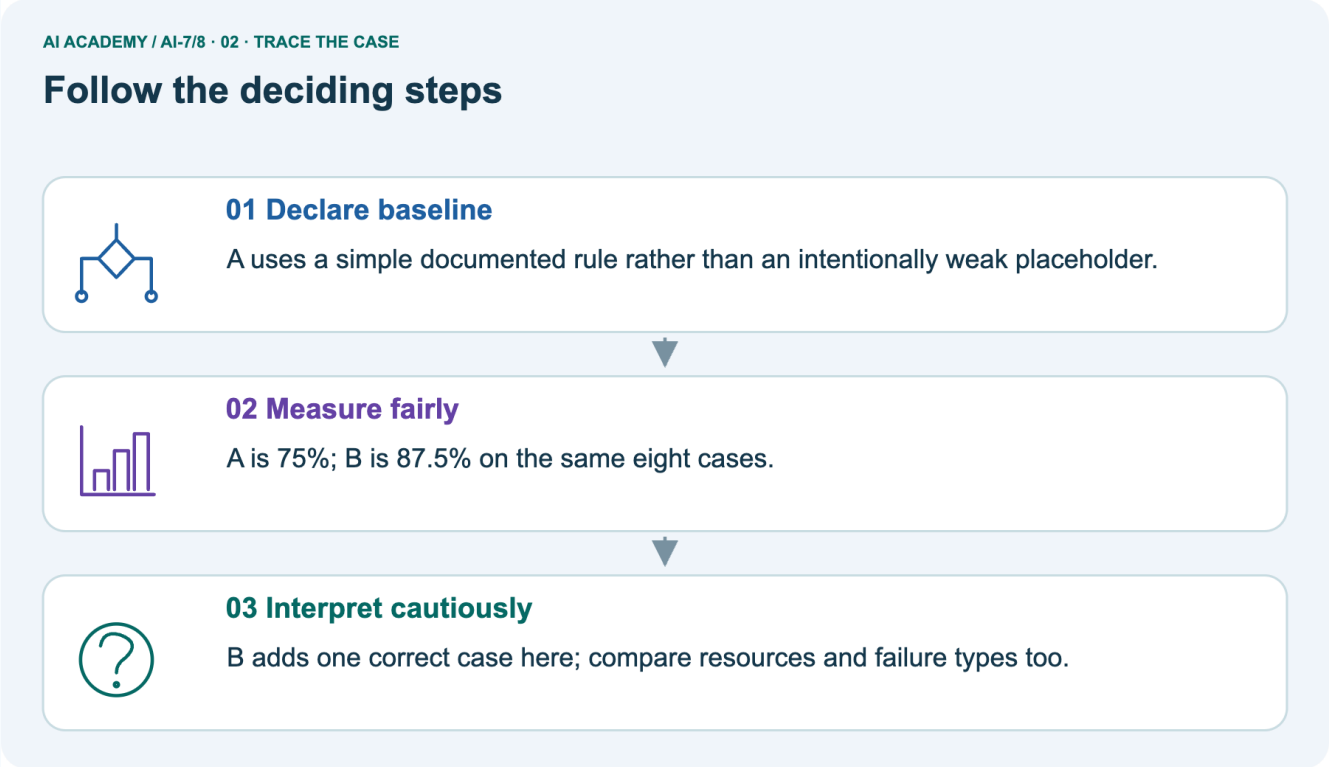
A **baseline** deserves a fair chance. Two classifiers answer the same eight fixed questions with the same allowed sources. Supplied facts: Simple rule A: 6/8 correct. New model B: 7/8 correct.

No test case is used to tune either version.

Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



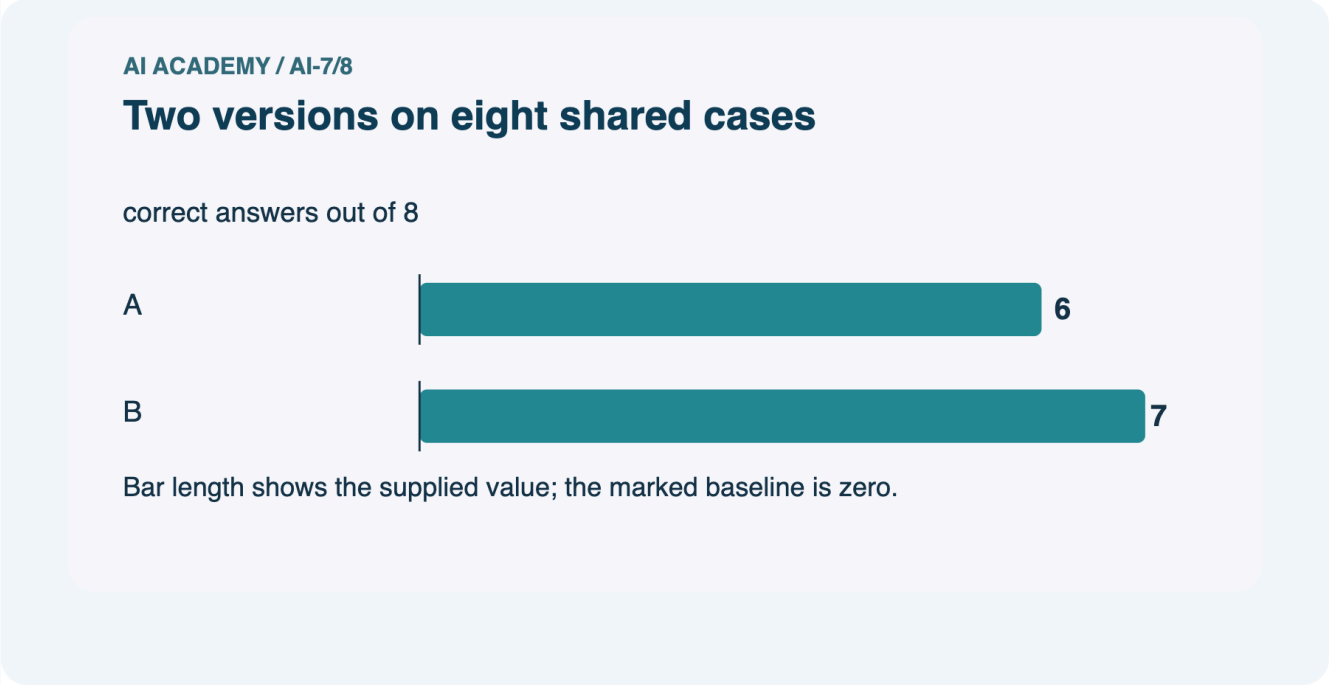
Follow the labelled picture with the matching explanation. Open the vector diagram

Declare **baseline**: A uses a simple documented rule rather than an intentionally weak placeholder. → Measure fairly: A is 75%; B is 87.5% on the same eight cases. → Interpret cautiously: B adds one correct case here; compare resources and failure types too.

Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Two versions on eight shared cases



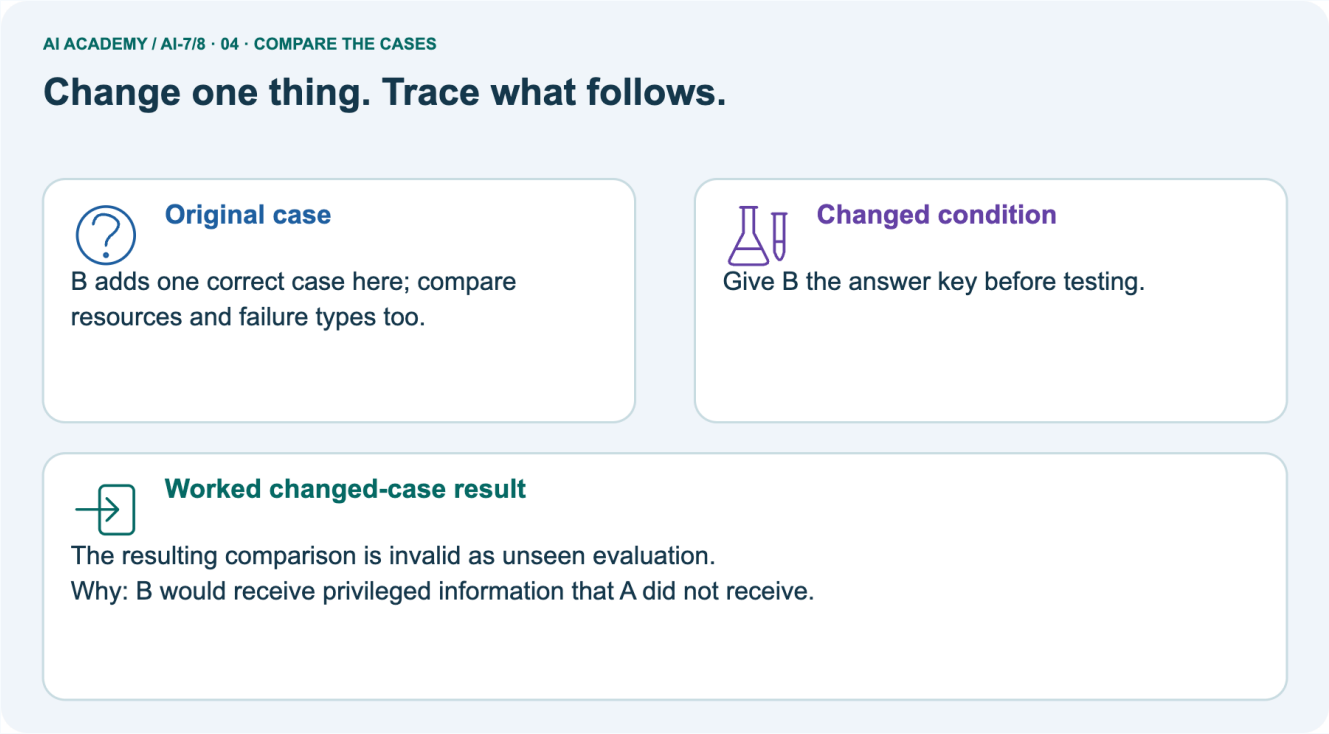
Follow the labelled picture with the matching explanation. Open the vector diagram

Two versions on eight shared cases. Units: correct answers out of 8. A: 6. B: 7.

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

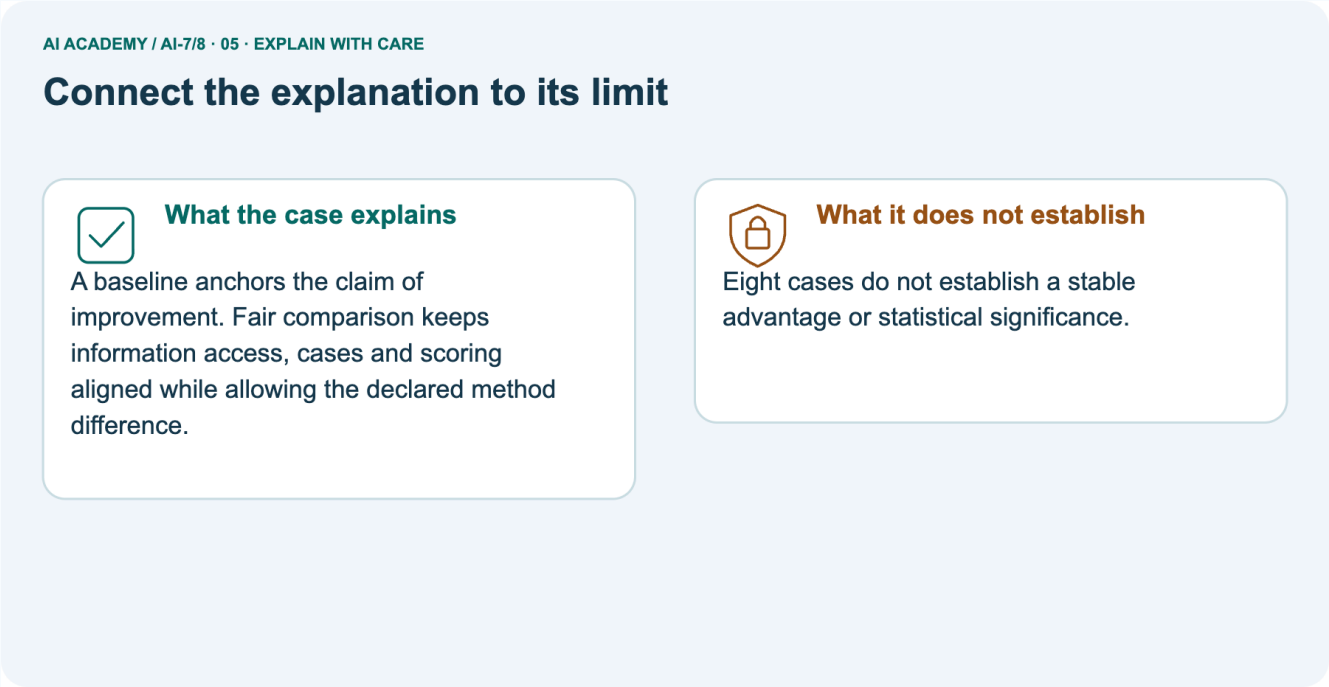
Original case: B adds one correct case here; compare resources and failure types too. Changed condition: Give B the answer key before testing. Expected result: The resulting comparison is invalid as unseen evaluation.

Reason: B would receive privileged information that A did not receive.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



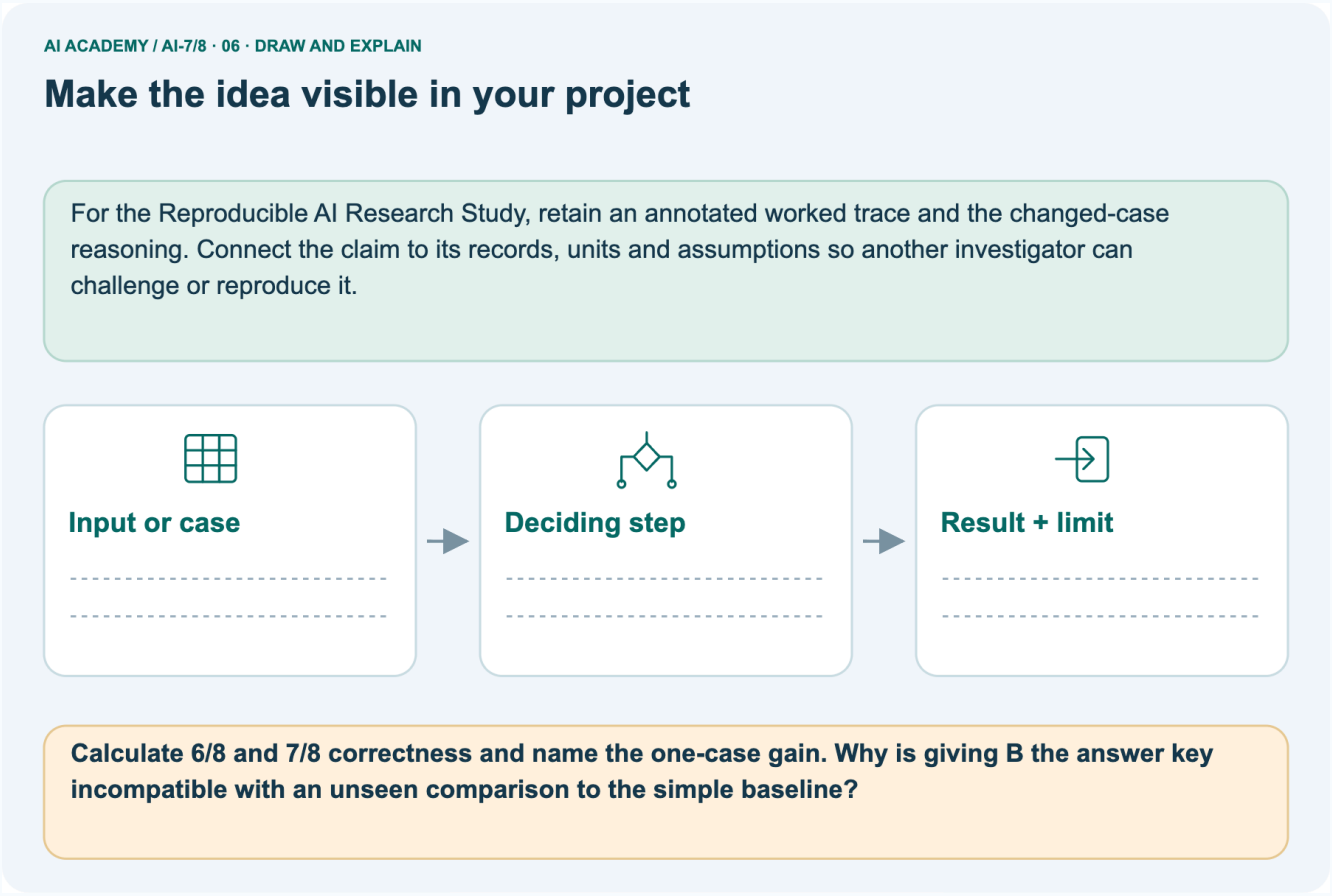
Follow the labelled picture with the matching explanation. Open the vector diagram

A **baseline** anchors the claim of improvement. Fair comparison keeps information access, cases and scoring aligned while allowing the declared method difference. Limit: These eight-case aggregate results alone establish neither a stable advantage nor a reported statistical-significance result.

Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

Draw your **input**, deciding step, result and limit. Supported question: Calculate 6/8 and 7/8 correctness and name the one-case gain. Why is giving B the answer key incompatible with an unseen comparison to the simple **baseline**?

Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

▼ Earlier wording, alternatives and picture transcripts

01. Baselines and fair comparisons: the case

02. Follow the deciding steps

03. Two versions on eight shared cases

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.